

# R.M.D. ENGINEERING COLLEGE

Smart India Hackathon 2026

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## PROJECT REPORT

# PRAMAAN

## Online Verification and Digital Certification System for Weighing and Measuring Instruments

**Problem Statement ID: SIH26036**

Software Development of an Online Verification System for Weighing and Measuring Instruments

Category: Software | Theme: Miscellaneous

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### Project Links

GitHub Repository: [github.com/haripriyasuresh072007-ai/HackCult-PraMaan](https://github.com/haripriyasuresh072007-ai/HackCult-PraMaan)

Live Prototype: [legal-metrology-eight.vercel.app](https://legal-metrology-eight.vercel.app)

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# 1. Introduction

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## a) Project Overview

Praamaan is a web-based online verification and digital certification system for weighing and measuring instruments used under India's Legal Metrology framework. It digitises the complete verification journey: instrument registration, online application for verification or re-verification, officer inspection and verification, issue of a QR-enabled digital certificate, and public verification of that certificate by scanning its QR code.

The platform is organised as role-based portals for Businesses, Legal Metrology Officers and Administrators, all working on one central record of instruments, applications and certificates. It is built as a full-stack Next.js application with a Prisma-managed PostgreSQL database and is designed for cloud deployment on Vercel.

Praamaan is a software prototype developed for the Smart India Hackathon. It is designed to support the work of Legal Metrology authorities and does not replace their statutory procedures or the judgement of the verifying officer. It is not an official government system.

**Project status:** Web-application prototype built around the core verification workflow, with a role-based dashboard. Production-readiness items such as stronger authentication, audit logs, data migration and a phased pilot are planned (Sections 6 and 10).

## b) Problem Statement

**Problem Statement ID:** SIH26036 (Ministry of Consumer Affairs, Food & Public Distribution, Department of Consumer Affairs)

**Title:** Software Development of an Online Verification System for Weighing and Measuring Instruments

- Under the Legal Metrology Act, 2009 and the Legal Metrology (General) Rules, 2011, weighing and measuring instruments used in trade must be periodically verified and stamped before they are put into use.
- Verification is carried out by Legal Metrology Officers of the State departments and by Government Approved Test Centres. It involves applications, scheduling, recording of observations, certificates, record keeping and monitoring of validity periods.
- These steps are largely manual, and records are often kept on paper or in isolated local systems. This causes delays and makes it hard to monitor verification status across jurisdictions.
- A unified online verification and digital certification system is needed to improve transparency, efficiency and ease of compliance.

## c) Objectives

- Digitise the complete verification process, from registration to certificate, instead of relying on paper-based steps.
- Enable online instrument registration and online applications for verification and re-verification.
- Provide officer inspection and verification workflows built around a digital checklist.
- Generate QR-enabled digital certificates with a unique certificate number and validity dates.
- Let anyone check the authenticity of a certificate instantly through its QR code.
- Provide centralised monitoring dashboards for businesses, officers and administrators.
- Build a foundation that can be extended to the full scope of SIH26036, including alerts, test centres, field support and integration.

## d) Target Users and Impact

Every business that uses a weighing or measuring instrument in trade must keep it verified, and each verification involves several parties. Pramaan is intended for the following users:

- **Businesses** using weighing and measuring instruments
- **Legal Metrology Officers** who inspect and verify instruments
- **Approved Testing Centres** that carry out verification work
- **Administrators and regulatory authorities** who monitor compliance
- **The public**, who can check a certificate's authenticity through its QR code

The expected benefits are a faster, paperless verification process, real-time tracking of applications and certificates, instant QR-based certificate verification and centralised monitoring of compliance. Beyond these, the project aims to improve transparency and accountability, reduce manual paperwork and processing time, and enable better monitoring of instrument validity. It also supports data-driven regulatory decisions, reduces the compliance burden for businesses, and builds public trust through verifiable digital certificates. The project aligns with UN Sustainable Development Goal 16 (Peace, Justice and Strong Institutions).

## e) Project Team

Table 1.1: Team details

| Item                 | Details                                                                   |
|----------------------|---------------------------------------------------------------------------|
| Team name            | HackCult_80                                                               |
| Institution          | R.M.D. Engineering College                                                |
| Problem Statement ID | SIH26036                                                                  |
| Theme and category   | Miscellaneous; Software                                                   |
| Team lead            | Haripriya S                                                               |
| Team members         | Jesika M V, Kodavala Dhurgasree, Rupendra M, Pranav Mani S, Sanjay Raaj D |
| Department and year  | Computer Science and Engineering, Year II                                 |
| Mentor               | Rajeshwari P                                                              |

## 2. Problem Background and Existing Solutions

### a) Current Practice and Challenges

Where verification runs on paper or in isolated local systems, four challenges shape the design of Pramaan:

- **Manual, paper-based processes.** Applications, inspections and certificates move on paper, which is slow.
- **Hard-to-track records.** Application status and inspection history are difficult to follow.
- **Validity tracking.** Re-verification is due at prescribed intervals, but there is no central view of valid certificates.
- **No simple public check.** A buyer has no easy way to confirm that a certificate is genuine.

The workload is also spreading: the 2025 amendments to the Government Approved Test Centre Rules widened the instrument categories that test centres can verify (18, later reported as 23), so shared digital records matter more.

### b) Research Landscape

Digital services for legal metrology already exist, and the team acknowledges this openly. The table below summarises the landscape and what it means for Pramaan.

Table 2.1: Existing work and its implication for this project

| Area                                      | What already exists                                                                                                                                                                       | Implication for Pramaan                                                                                                             |
|-------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------|
| <b>National and state portals</b>         | State governments run their own portals for licences, registration and verification. The Department of Consumer Affairs, with NIC, has announced the national eMaap portal to unify them. | Online registration alone is not novel. Pramaan focuses on the verification workflow and is designed to work with existing systems. |
| <b>Government Approved Test Centres</b>   | The GATC Rules, 2013 were amended in 2025 to cover more instrument categories, with privately operated centres verifying instruments.                                                     | Verification work is shared between State officers and test centres, so records need to be shared as well.                          |
| <b>QR-based verification transparency</b> | An OIML Bulletin article (2025) describes QR codes on verified instruments that show verification details without a login (example: Austria's BEV).                                       | QR verification is established international practice. Pramaan applies it to certificates for instant public checks.                |
| <b>Legal framework</b>                    | The Legal Metrology Act, 2009 and the General Rules, 2011 require periodic verification, stamping and re-verification of instruments.                                                     | Praamaan digitises the procedure and does not change the legal requirements.                                                        |

### c) What Sets Pramaan Apart

Online registration and digital certificates are not new in themselves. The difference lies in how the whole workflow is connected:

- **End-to-end digital workflow.** Registration to QR check runs in one system.
- **Instant QR verification.** Anyone can confirm a certificate's authenticity by scanning its QR code.
- **Role-based ecosystem.** Businesses, officers, administrators and the public each see what they need.
- **Traceable instrument record.** Each instrument keeps a digital history.
- **Honest scope.** The prototype status, planned features and limits are stated clearly throughout.

### 3. Proposed Solution

#### a) Solution Overview

A business owner registers on Pramaan and adds each weighing or measuring instrument with its type, manufacturer, model, serial number and capacity. The owner then applies online for verification or re-verification and chooses the inspection date and location. The application enters an officer's queue, an officer is assigned, and the inspection is carried out against a digital checklist.

Once the instrument is verified against the applicable standards, Pramaan issues a digital certificate with a unique number, verification and validity dates, and a QR code. Anyone can scan the QR code to check that the certificate is genuine. Dashboards give each role a live view of instruments, applications and certificates.

**Important honest point:** Pramaan is a demonstration prototype built for SIH26036. It is not an official government system, it does not replace verification by authorised officers, and it has not been connected to State portals or the national eMaap portal.

#### b) Verification Workflow

Table 3.1: Workflow steps

| Step | Action                                                                                                      |
|------|-------------------------------------------------------------------------------------------------------------|
| 1    | Register users: business owners, officers and administrators, with role-based access.                       |
| 2    | Register the instrument: type, manufacturer and model, serial number and capacity.                          |
| 3    | Apply online: confirm the instrument type, choose the inspection date and location, and submit the request. |
| 4    | The application enters the queue and an officer is assigned for inspection.                                 |
| 5    | The officer carries out the inspection using the digital inspection checklist.                              |
| 6    | The officer verifies the instrument against standards and validates its details.                            |
| 7    | Pramaan issues the certificate: unique number, validity dates and QR code.                                  |
| 8    | Anyone scans the QR code to check the certificate's authenticity publicly.                                  |
| 9    | Dashboards track status and validity, and the record supports the next re-verification cycle.               |

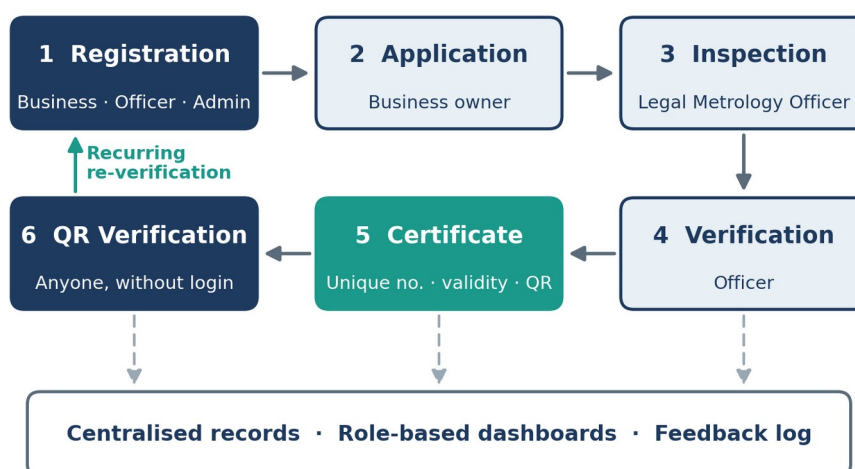


Figure 3.1: Pramaan verification workflow and the role at each stage

### c) System Architecture

The current implementation is an integrated Next.js application. The web interface, application logic, data access and QR verification run within one application, which is deployed on Vercel.

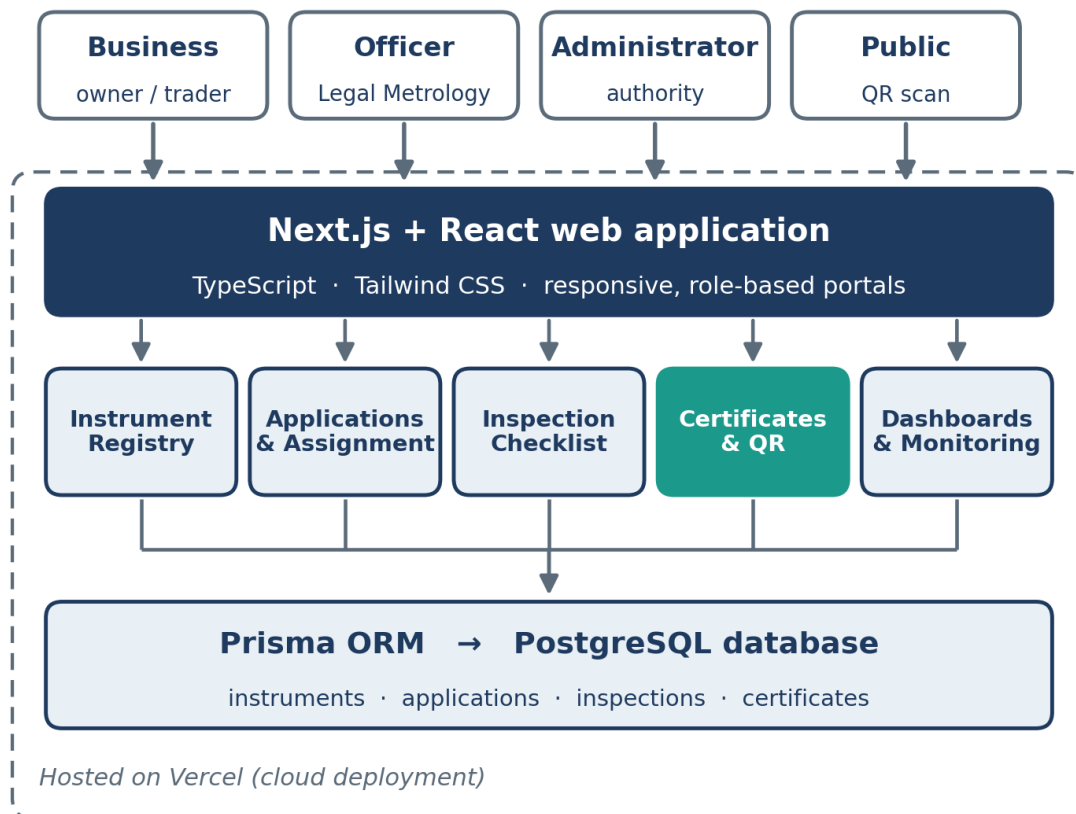


Figure 3.2: Pramaan system architecture (integrated Next.js application)

### d) Design Principle: One Central Record

All portals read and write the same central records through a single data layer (Prisma). Each role sees only the screens and actions meant for it, and the public QR check is read-only. Because every action updates one record, dashboards, certificates and verification results stay consistent with each other.

## 4. System Modules and Technology Stack

### a) Modules

Table 4.1: Modules and who uses them

| Module                      | What it does                                                                                             | Used by                 |
|-----------------------------|----------------------------------------------------------------------------------------------------------|-------------------------|
| Access and roles            | Registration, login and separate portals for Business, Officer and Administrator with role-based access  | All users               |
| Instrument registry         | Stores instrument type, manufacturer and model, serial number and capacity                               | Business, Administrator |
| Application management      | Online verification and re-verification requests with inspection date and location, plus status tracking | Business, Officer       |
| Queue and assignment        | Application queue and assignment of an officer for inspection                                            | Officer, Administrator  |
| Inspection and verification | Digital inspection checklist, cross-check against standards and validation of instrument details         | Officer                 |
| Certificate service         | Unique certificate number, verification and validity dates, and a QR-enabled certificate                 | Officer, Business       |
| QR verification             | Public scan-and-check of a certificate's authenticity                                                    | Public, Business        |
| Dashboards and monitoring   | Central view of instruments, applications, processing status, certificates and system health             | Business, Administrator |

### b) Technology Stack

Table 4.2: Technologies used

| Technology   | Purpose                          |
|--------------|----------------------------------|
| Next.js      | Full-stack web application       |
| React        | Interactive user interfaces      |
| TypeScript   | Type-safe development            |
| Prisma       | Database ORM and management      |
| PostgreSQL   | Application database             |
| Tailwind CSS | Responsive UI styling            |
| QR code      | Digital certificate verification |
| Vercel       | Cloud deployment and hosting     |

The stack uses open-source, widely supported tools, which keeps licensing costs low and makes the project easy for other developers to maintain.

### c) Core Features

- Role-based portals for Business, Officer and Administrator
- Instrument registration with type, manufacturer, model, serial number and capacity
- Online application for verification and re-verification, with inspection date and location
- Application queue and officer assignment

- Digital inspection checklist and verification against standards
- Unique certificate number with verification and validity dates
- QR-enabled digital certificates and public QR verification
- Dashboards for instruments, applications, processing status and certificates, with system-health indicators

#### d) Prototype Interface

Figure 4.1 shows the prototype dashboard of the Business portal. The left menu gives access to Dashboard, Instruments, Applications, Certificates and Profile. Summary cards show registered instruments, applications, applications under processing and certificates issued. Recent applications carry status labels such as Pending, Under Review and Verified, and a system-health panel shows the state of the application service, officer workflow, certificate service and QR verification. The values shown are prototype data.

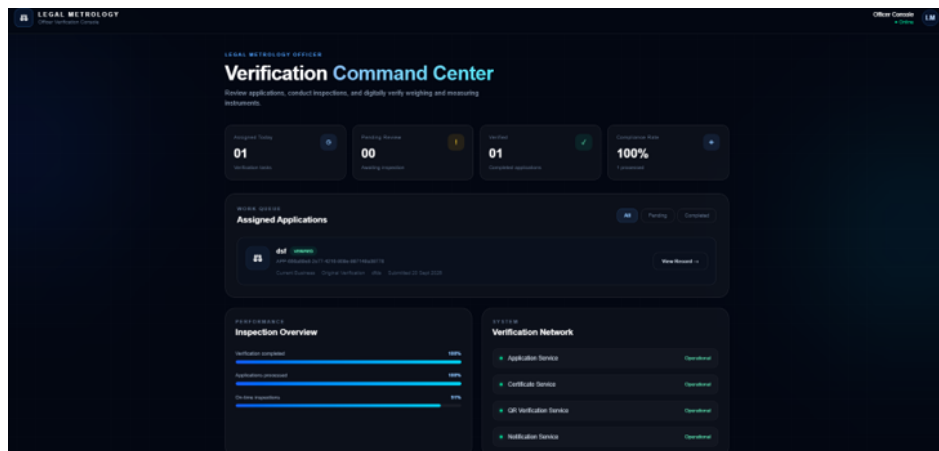


Figure 4.1: Prototype dashboard of the Pramaan Business portal

## 5. Roles, Records and Certificate Verification

### a) Roles and Access

Table 5.1: Roles and what each can do

| Role                           | What the role can do                                                                                                                 |
|--------------------------------|--------------------------------------------------------------------------------------------------------------------------------------|
| <b>Business owner</b>          | Register instruments, apply for verification or re-verification, track application status and view certificates                      |
| <b>Legal Metrology Officer</b> | Work the assigned application queue, carry out the inspection with the digital checklist, and verify instrument details and accuracy |
| <b>Administrator</b>           | Monitor applications and certificates across the system through the admin dashboard, and apply manual overrides where needed         |
| <b>Public verifier</b>         | Scan a certificate's QR code to check its authenticity without logging in                                                            |

### b) Core Records

Four kinds of record connect the workflow. Each one links back to the instrument, which is what gives every instrument a traceable history.

Table 5.2: Core records

| Record             | Key details                                                                                                                                    |
|--------------------|------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Instrument</b>  | Instrument type, manufacturer and model, serial number, capacity                                                                               |
| <b>Application</b> | Instrument, request type (verification or re-verification), inspection date and location, and status such as Pending, Under Review or Verified |
| <b>Inspection</b>  | Assigned officer and the results of the digital inspection checklist                                                                           |
| <b>Certificate</b> | Unique certificate number, verification date, validity date and QR code                                                                        |

### c) QR Certificate Verification

Each certificate carries a QR code. Scanning it opens a public verification page that checks the certificate against the central record and shows its details, including the validity dates. No login is needed and the page is read-only.

Checking against the source record, and not the printed copy, is what makes a forged or altered paper certificate easy to catch. This follows the approach described in the OIML bulletin on digital transparency in legal metrology, where verification data can be read by scanning a QR code without authentication.

## 6. Feasibility, Strategy and Challenges

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### a) Technical and Operational Feasibility

- Uses open-source technology: Next.js, React, Tailwind CSS, PostgreSQL and Prisma.
- Scalable architecture for a growing user base.
- Designed to integrate with existing systems.
- The administrator dashboard allows manual overrides and monitoring.

### b) Economic Feasibility

- Minimal software licensing costs.
- Lightweight cloud infrastructure, with Vercel providing cost-effective deployment.
- Reduces paperwork and manual processing.

### c) Implementation Strategy

- **Phased deployment.** Start with selected users and expand gradually.
- **User training.** Provide simple dashboards and guidance for officers and businesses.
- **Secure the platform.** Add authentication, role-based access and audit logs.

### d) Challenges and Response

Table 6.1: Main challenges and the planned response

| Challenge             | Why it matters                                                                         | Planned response                                      |
|-----------------------|----------------------------------------------------------------------------------------|-------------------------------------------------------|
| <b>Data migration</b> | Existing records sit on paper or in local systems and must move to the digital system. | Start with selected users and expand gradually.       |
| <b>User adoption</b>  | Officers and businesses need basic training to use the platform.                       | Simple dashboards, guidance and user training.        |
| <b>Security</b>       | Stronger authentication is required before production use.                             | Add authentication, role-based access and audit logs. |

## 7. Compliance and Project Status

### a) Mapping to Problem Statement SIH26036

The table maps the requirements listed in the problem statement to what Pramaan covers. **Core** means part of the core solution defined for this project, **Partial** means covered in part, and **Planned** means not in the current version and part of the roadmap. Statuses describe the scope defined in the project's idea submission and will be updated as the build progresses.

Table 7.1: Problem statement requirements and current status

| Requirement                                                        | Current implementation                                                                                                                 | Status  |
|--------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------|---------|
| <b>Online registration and profile management of stakeholders</b>  | Role-based registration and secure profile creation for Business, Officer and Administrator. Test centres are not yet a separate role. | Partial |
| <b>Applications for verification and re-verification</b>           | Online application with instrument type, inspection date and location                                                                  | Core    |
| <b>Scheduling and allocation of verification work</b>              | Application queue and officer assignment. Allocation to test centres is not yet included.                                              | Partial |
| <b>Digital recording of inspection observations and results</b>    | Digital inspection checklist and verification result                                                                                   | Core    |
| <b>QR-enabled digital verification certificates</b>                | Unique certificate number, verification and validity dates, QR code                                                                    | Core    |
| <b>Digital repository of certificates and instrument records</b>   | Central records for instruments and certificates                                                                                       | Core    |
| <b>Tracking of validity and re-verification due dates</b>          | Validity dates are stored on certificates. Due-date views are planned.                                                                 | Partial |
| <b>Alerts and reminders for expiring validity</b>                  | Not in the current version                                                                                                             | Planned |
| <b>Dashboards for status, pendency and enforcement</b>             | Dashboards for instruments, applications, processing status and certificates. Pendency and enforcement views are planned.              | Partial |
| <b>Search and retrieval of records and certificates</b>            | Public QR verification of a certificate. Full record search is planned.                                                                | Partial |
| <b>Role-based secure login</b>                                     | Role-based access. Stronger authentication and audit logs are planned.                                                                 | Partial |
| <b>Upload of photographs and supporting documents</b>              | Not in the current version                                                                                                             | Planned |
| <b>Export and printing of certificates and reports</b>             | Not in the current version                                                                                                             | Planned |
| <b>Mobile support for field verification</b>                       | Responsive web interface. A dedicated field workflow is planned.                                                                       | Partial |
| <b>Technical documentation: architecture, security, deployment</b> | This report (Sections 3 and 8)                                                                                                         | Core    |

## b) Scope and Roadmap Checklist

The table below separates the core workflow from the items planned as future work.

Table 7.2: Core scope and planned items

| Item                                                                         | Status  |
|------------------------------------------------------------------------------|---------|
| Role-based portals for Business, Officer and Administrator                   | Core    |
| Instrument registration                                                      | Core    |
| Online application for verification and re-verification                      | Core    |
| Application queue and officer assignment                                     | Core    |
| Digital inspection checklist and verification                                | Core    |
| Unique certificate number and QR-enabled digital certificate                 | Core    |
| Public QR verification of certificates                                       | Core    |
| Dashboards and system-health view                                            | Core    |
| Cloud deployment on Vercel with a Prisma-managed PostgreSQL database         | Core    |
| Production-grade authentication and audit logs                               | Planned |
| Migration of existing paper and local records                                | Planned |
| Phased pilot with selected users and user training                           | Planned |
| Test centre (GATC) role and allocation of verification work                  | Planned |
| Expiry alerts and reminders                                                  | Planned |
| Photograph and document upload, and certificate and report export            | Planned |
| Mobile workflow for field officers                                           | Planned |
| Pendency and enforcement dashboards                                          | Planned |
| Integration with State Legal Metrology portals and the national eMaap portal | Planned |
| Field validation with Legal Metrology departments and test centres           | Planned |

## c) Progress Statement

**Status:** Web-application prototype built around the core verification workflow, from instrument registration to public QR verification. Production hardening, stakeholder expansion and integration with existing systems are future work.

## 8. Deployment and Demonstration

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### a) Deployment Approach

- Pramaan is a Next.js application. Vercel is the deployment target for cloud hosting, chosen for its low cost.
- Prisma manages the PostgreSQL database used by the application.
- Rollout is phased: start with selected businesses and officers, gather feedback and expand gradually.
- A production deployment would need stronger authentication (Section 10).
- Integration with existing State and national systems would follow the pilot, subject to access from the authorities.

### b) Security Framework

- Role-based access control separates the Business, Officer and Administrator portals.
- Authentication protects every portal. Stronger authentication is required before production use.
- Audit logs of officer and administrator actions are planned.
- Certificates are checked against the central record, and the public QR page is read-only.
- The administrator dashboard allows monitoring and manual overrides.

### c) Demo Walkthrough

- Sign in as a Business owner, open the dashboard and register an instrument.
- Submit an application for verification or re-verification with an inspection date and location.
- Sign in as an Officer, open the application queue and complete the digital inspection checklist.
- Verify the instrument and issue the certificate with its unique number, validity dates and QR code.
- Scan the QR code to show public verification of the certificate.
- Show the Administrator dashboard for monitoring.
- Backup plan: if the live demo is unavailable, use a recorded video or screenshots.

## 9. Limitations and Disclaimer

Praamaan is a software prototype developed for demonstration under SIH26036. It is not an official government system and does not replace verification by authorised Legal Metrology Officers or Government Approved Test Centres. In particular:

- Stronger authentication is required before any production use.
- Integration with State Legal Metrology portals and the national eMaap portal is a design goal and is not part of the current version.
- Verification standards, validity periods and fees come from the Legal Metrology rules and vary by instrument category. The platform must be aligned with the current rules and validated with the authorities before operational use.
- The legal standing of digital certificates depends on the rules and notifications of the competent authority.
- Test centres, expiry alerts, document upload, certificate and report export, and a dedicated field workflow are not yet included.
- The figures shown in the prototype dashboard are prototype data and do not represent real instruments or certificates.
- The system has not yet been tested in the field with officers, test centres or businesses.
- Verification results must be confirmed by an authorised Legal Metrology Officer before a certificate is relied on.

## 10. Future Work

The current prototype covers the core verification workflow. The extensions below make up the remaining roadmap, and none of them is part of the current version.

Table 10.1: Planned enhancements

| Enhancement                        | What will be done                                                                                                          | Limitation addressed                                 |
|------------------------------------|----------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------|
| <b>Production-grade security</b>   | Add stronger authentication, session security and audit logs for officer and administrator actions.                        | Authentication needs strengthening                   |
| <b>Test centre (GATC) support</b>  | Add Government Approved Test Centres as a separate role, with allocation of verification work to officers or test centres. | Roles limited to Business, Officer and Administrator |
| <b>Expiry alerts and reminders</b> | Send automated reminders before a verification's validity ends and show due dates on dashboards.                           | No automated alerts                                  |
| <b>Attachments</b>                 | Allow photographs and supporting documents to be attached to applications and inspections.                                 | No document upload                                   |
| <b>Export and printing</b>         | Produce printable and exportable certificates and reports.                                                                 | No export facility                                   |
| <b>Search and retrieval</b>        | Search records and certificates by certificate number, instrument serial number, business and date.                        | Only QR-based lookup                                 |
| <b>Detailed inspection records</b> | Record measured observations and results in detail, not only checklist outcomes.                                           | Checklist-level recording                            |
| <b>Field and mobile workflow</b>   | Support inspections on mobile devices in the field, including working with poor connectivity.                              | Web-only workflow                                    |

| Enhancement                                | What will be done                                                                                                 | Limitation addressed       |
|--------------------------------------------|-------------------------------------------------------------------------------------------------------------------|----------------------------|
| <b>Pendency and enforcement dashboards</b> | Add views of pending work and enforcement activity across jurisdictions.                                          | Basic status dashboards    |
| <b>Data migration tools</b>                | Import existing paper and local records into the central system.                                                  | Records outside the system |
| <b>Integration with existing systems</b>   | Connect with State Legal Metrology portals and the national eMaap portal, subject to access from the authorities. | Standalone system          |
| <b>Pilot and training</b>                  | Run a phased pilot with selected businesses and officers, with simple guides and training.                        | User adoption              |
| <b>Field validation</b>                    | Validate with Legal Metrology departments, test centres and businesses.                                           | No operational validation  |

## 11. Conclusion

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Pramaan shows how the Legal Metrology verification journey can be delivered as one connected digital workflow. It brings instrument registration, online applications, officer inspection, verification, QR-enabled digital certificates and public QR verification together in role-based portals on a single central record, built with widely used open-source tools.

The project is honest about its scope. It is a prototype: authentication must be strengthened, existing records must be migrated, test centres, alerts, attachments and field support are still to be added, and integration with existing State and national portals depends on the authorities. The main next steps are production-grade security, a phased pilot with selected users and training, and the additions listed in Table 7.2.

In one sentence: Pramaan is a role-based, QR-enabled online verification prototype for weighing and measuring instruments, with the core workflow from registration to public verification in place and production hardening, stakeholder expansion and integration left as future work.

## 12. References

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### Government and regulatory sources

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- Weighing News, “Consumer affairs ministry amends legal metrology rules”, May 2026.

### Tools and software

- Next.js, React, TypeScript, Prisma, PostgreSQL, Tailwind CSS, QR code generation and Vercel.